

4.7.6 If the films show classic changes suggestive of osteomyelitis (cortical erosion, periosteal reaction, mixed lucency, and sclerosis), and if there is little likelihood of neuro-osteopathy, it is reasonable to initiate treatment for presumptive osteomyelitis, preferably after obtaining appropriate specimens for culture. IDSA 2012 guidelines (Adopted)

4.7.7 If the Diabetic Foot care centre suspects acute Charcot arthropathy, offer treatment with a non-removable off-loading device. Only consider treatment with a removable off-loading device if a non-removable device is not advisable because of the clinical or the person's circumstances. NICE clinical guideline NG19 (Adapted)

4.7.8 Do not offer bisphosphonates to treat acute Charcot arthropathy, unless as part of a clinical trial. NICE clinical guideline NG19 (Adopted)

4.7.9 Monitor the treatment of acute Charcot arthropathy using clinical assessment. This should include measuring foot–skin temperature difference and taking serial X-rays until the acute Charcot arthropathy resolves. Acute Charcot arthropathy is likely to resolve when there is a sustained temperature difference of less than two degrees centigrade between both feet and when X-ray changes show no further progression. NICE clinical guideline NG19 (Adapted)

4.7.10 People who have a foot deformity that may be the result of a previous Charcot arthropathy are at high risk of ulceration and should be cared for by the Diabetic Foot care centre. NICE clinical guideline NG19 (Adopted)

4.7.10 The Diabetic Foot care centre should care for people who have a foot deformity resulting from a previous Charcot's arthropathy as they are at high risk of ulceration. NICE clinical guideline NG19 (Adapted)

4.8: SURGICAL INTERVENTIONS AND REVASCULARIZATION

4.8.1 Consult a surgical specialist in all cases of diabetic foot infections that are moderate or severe. IWGDF 2015 (Adapted)

4.8.2 Perform urgent surgical intervention in cases of deep abscesses, compartment syndrome and necrotizing soft tissue infections. IWGDF 2015 (Adapted)

4.8.3 Debride any wound that has necrotic tissue or surrounding callus; the required procedure may range from minor to extensive. IWGDF 2015 (Adapted)

4.8.4 Perform urgent surgical intervention for most foot infections accompanied by gas in the deeper tissues, an abscess, or necrotizing fasciitis, and less urgent surgery for wounds with substantial nonviable tissue or extensive bone or joint involvement. IDSA 2012 guidelines (Adapted)

Additional note: In those with a non-severe infection, carefully observing the effectiveness of medical therapy and the demarcation line between necrotic and viable tissue before operating may be prudent.

4.8.5 Consider surgical intervention in cases of osteomyelitis accompanied by: spreading soft tissue infection; destroyed soft tissue envelope; progressive bone destruction on X-ray; or, bone protruding through the ulcer. IWGDF 2015 (Adapted)

4.8.6 Remove slough, necrotic tissue & surrounding callus with sharp debridement in preference to other methods, taking relative contraindications such as severe ischemia into account. IWGDF 2015 (Adapted)

4.8.7 Consider digital flexor tenotomy to prevent a toe ulcer when conservative treatment fails in a high-risk patient with diabetes, hammertoes and either a pre-ulcerative sign or an ulcer on the toe. (weak; low) IWGDF 2015 (Adopted)

4.8.8 Consider Achilles tendon lengthening, joint arthroplasty, single or pan metatarsal head resection or osteotomy to prevent a recurrent foot ulcer when conservative treatment fails in a high-risk patient with diabetes and a plantar foot ulcer. (weak; low) IWGDF 2015 (Adopted)

Management of Peripheral Artery Disease in patients with Diabetic foot problems

4.8.9 Examine a patient with diabetes annually for the presence of peripheral artery disease (PAD); this should include, at a minimum, taking a history and palpating foot pulses. (Strong; low) IWGDF 2015 (Adopted)

4.8.10 Evaluate a patient with diabetes and a foot ulcer for the presence of PAD. Determine, as part of this examination, ankle or pedal Doppler arterial waveforms; measure both ankle systolic pressure and systolic ankle brachial index (ABI). (Strong; Low) IWGDF 2015 (Adopted)

4.8.11 Use bedside non-invasive tests to exclude PAD. No single modality has been shown to be optimal. Measuring ABI (with <0.9 considered abnormal) is useful for the detection of PAD. Tests that largely exclude PAD are the presence of ABI $0.9-1.3$, toe brachial index (TBI) ≥ 0.75 and the presence of triphasic pedal Doppler arterial waveforms. IWGDF 2015 (Adapted)

4.8.12 In patients with a non-healing ulcer with either an ankle pressure $<50\text{mmHg}$ or ABI <0.5 consider urgent vascular imaging and revascularisation. (Strong; Low) IWGDF 2015 (Adopted)

4.8.13 Consider vascular imaging and revascularisation in all patients with a foot ulcer in diabetes and PAD, irrespective of the results of bedside tests, when the ulcer does not improve within 6 weeks despite optimal management. (Strong; Low) IWGDF 2015 (Adopted)

4.8.14 Do not consider Diabetic microangiopathy to be the cause of poor wound healing in patients with a foot ulcer. IWGDF 2015 (Adapted)

4.8.15 To obtain anatomical information when revascularisation is being considered, use one of these tests - Colour Doppler ultrasound, CT-angiography, MR-angiography or intra-arterial digital subtraction angiography. Evaluate the entire lower extremity arterial circulation, with detailed visualization of below-the-knee and pedal arteries. IWGDF 2015 (Adapted)

4.8.16 Offer duplex ultrasound as first-line imaging to all people with peripheral arterial disease for whom revascularization is being considered. Take the decision of revascularisation on the basis of colour doppler findings and use DSA for defining the vascular anatomy prior to the procedure. (Adapted from NICE 2012 PAD Guidelines, last modified August 2015)

4.8.17 The aim of revascularisation is to restore direct flow to at least one of the foot arteries, preferably the artery that supplies the anatomical region of the wound, and adequate revascularization should be assessed post-operatively with a colour Doppler wave-fronts (preferable) or a hand held Doppler probe used bedside. IWGDF 2015. (Adapted)

4.8.18 A centre treating patients with a foot ulcer in diabetes should have liaison / association with a centre having the expertise necessary to diagnose and treat PAD; both endovascular techniques and bypass surgery should be available. (Adapted) IWGDF 2015

4.8.19 The multidisciplinary team should treat the patient after a revascularisation procedure for a foot ulcer in diabetes, as part of a comprehensive care plan. (Strong; Low) IWGDF 2015 (Adapted).

4.8.20 There is inadequate evidence to establish which revascularisation technique is superior and a multidisciplinary team should decide the technique of revascularization for a patient based on a number of individual factors, such as morphological distribution of PAD, availability of autogenous vein, patient co-morbidities and local expertise. (Strong; Low) IWGDF 2015 (Adapted)

4.8.21 Give emergency treatment to patients with signs of PAD and a foot infection as they are at particularly high risk for major limb amputation. IWGDF 2015 (Adapted)

4.8.22 Avoid revascularisation in patients in whom, from the patient perspective, the risk-benefit ratio for the probability of success is unfavourable*. (Strong;Low) IWGDF 2015 (Adopted)

Explanatory note: Unfavourable risk benefit ratio would indicate those patients who are frail, elderly, bed ridden, having low life expectancy, multiple co-morbidities imposing high risk for surgical intervention, etc

4.8.23 All patients with diabetes and an ischemic foot ulcer should receive aggressive cardiovascular risk management including support for cessation of smoking, treatment of hypertension and prescription of a statin as well as low-dose aspirin or clopidogrel. (Strong;Low) IWGDF 2015 (Adopted)

4.8.24 Do not offer major amputation to people with critical limb ischaemia unless all options for revascularisation have been considered by a vascular multidisciplinary team. Major amputation without giving a chance for revascularization is indicated only in lifesaving situations like foot causing septicemia, wet gangrene, or completely destroyed foot (post charcot's or osteomyelitis etc). (Adapted) NICE 2012 PAD guidelines, last modified August 2015

THE WAY FORWARD FOR INDIA